

Allie (Minh-Anh) Nguyen

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EDUCATION

University of California, Berkeley

Ph.D. Candidate, Environmental Engineering

Master of Science, Environmental Engineering, GPA: 4.0/4.0

Berkeley, CA

Aug 2022 – May 2027 (Expected)

Dec 2023

University of Minnesota - Twin Cities

Bachelor of Environmental Engineering, with Distinction, GPA: 3.81/4.0

Minneapolis, MN

May 2022

SKILLS

Laboratory: Digital PCR, DNA/RNA extraction from water and wastewater, UV-Vis spectroscopy, cell culture, sequencing batch reactor operation, chemical and microbial analysis of water quality.

Software: R/RStudio, LaTeX, MATLAB, Microsoft Office.

Languages: Vietnamese (native speaker proficiency), English (professional proficiency).

Certifications: Google Data Analytics Professional Certificate (via Coursera), Master Certified Tutor (CRLA Level 3).

RESEARCH EXPERIENCE

Graduate Student Researcher

Aug 2022 – Present

University of California, Berkeley, Department of Civil and Environmental Engineering

Berkeley, CA

- Design and execute novel experiments to study how surfactants influence the decay of viral nucleic acid signals in wastewater across 8 human respiratory viruses.
- Conduct extensive data analysis on San Francisco sub-sewersheds' COVID-19 wastewater and case datasets to evaluate how neighborhood-scale sampling can improve health equity in wastewater-based surveillance.
- Develop protocols, lead experiments, and contribute in building experimental models to investigate the microbial quality of advanced treated water blended with conventional surface water in pilot distribution systems.
- Mentor 4 master's students on microbial experimental techniques.

Undergraduate Research Assistant

Jan 2021 – Aug 2022

University of Minnesota Twin Cities, Department of Civil, Environmental, and Geo- Engineering

- **Biochar for Nutrient Removal from Wastewater:** Characterized ammonia sorption isotherms to biochars and DAX-8 using segmented flow analysis and UV-Vis spectroscopy across six biochar-amended sequencing batch reactors.
- **Hydraulics of Sea Lamprey Barriers:** Evaluated sea lamprey barrier efficacy through fluid mechanics experiments with model fixed-crest barriers in hydraulic flumes and analysis of flow velocities, lamprey detachment forces, and turbulent kinetic energy.
- **Biomethane Generation from Municipal Waste:** Measured anaerobic digestion capability of organic waste in Hanoi, Vietnam via bench-scale biochemical methane potential tests.

MENTORING EXPERIENCE

Graduate Student Instructor

Jan 2025 – May 2025

University of California, Berkeley, Department of Civil and Environmental Engineering

Berkeley, CA

Course: Environmental Biological Processes (CE 211B)

- Collaborated with course instructor to revise assignments and develop grading rubrics.
- Graded weekly homework assignments and exams of 35 graduate students.
- Held weekly office hours and led exam review sessions to reinforce key concepts, provide example problems, and enhance student comprehension.

Undergraduate Student Instructor

Sep 2019 – May 2022

University of Minnesota Twin Cities

Minneapolis, MN

Courses: Computer Applications (CEGE 3101); Introduction to Environmental Engineering (CEGE 3501)

- Graded assignments and collaborated with course instructor to develop grading rubrics for 60 upper-division undergraduate students in the Department of Civil, Environmental, and Geo- Engineering.

Course: First-Year Experience (CSE 1001)

- Prepared teaching materials, held lectures, and managed class logistics for 30 first-year undergraduate students in the College of Science and Engineering.

SURF Graduate Student Mentor

Jan 2023 – Aug 2023

University of California, Berkeley, Office of Undergraduate Research

Berkeley, CA

- Led draft proposal workshops and weekly office hours for applicants of the Summer Undergraduate Research Fellows (SURF) program.
- Hosted summer workshops on research skills with an attendance of 50 SURF fellows.
- Organized and facilitated the SURF program's orientation, weekly group meetings, and students' final presentations.

LEADERSHIP

Beyond Academia (University of California, Berkeley)

Apr 2023 – Present

Logistics Co-director (May 2024 – Present); **Finance Co-director** (Jan 2024 – May 2024)

- Oversee logistics and \$70,000+ budget for all events and operations. Managed logistics, recruited guest speakers, and moderated panels for the 2024 and 2025 Beyond Academia conferences (650-1300+ attendees each).
- Lead 10+ graduate student volunteers and manage external partnerships with UC Berkeley Graduate Division, Graduate Professional Development, and Berkeley Career Engagement.

Graduate Women of Engineering (University of California, Berkeley)

Sep 2022 – Present

Co-President (Jun 2024 – Present); **Career and Professional Development Committee Head** (Jun 2023 – Jun 2024)

- Lead and oversee operations, officer meetings, external and internal communications, elections, and administration compliance.
- Manage external partnerships and communications with faculty, campus offices, and partner student organizations.
- Coordinate strategic initiatives and events, including newsletters, inter-organization collaborations, and external professional opportunities.
- Organized the annual Career Panel and Speaker Series for women-identifying engineering graduate students at UC Berkeley.